

KALIANGET MADURA ISLAND, IndonesiaWMO: **969730**Lat: **7.0413S**Lon: **113.9156E**Elev: **10**StdP: **14.69**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	74.6	75.4	67.7	101.8	81.2	68.9	106.3	81.1	32.6	82.9	29.0	83.8	2.1	300	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	10.8	92.0	81.3	91.2	81.1	90.4	80.9	84.1	89.8	83.2	89.1	82.5	88.6	10.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.4	169.1	89.1	81.4	163.5	88.2	80.7	159.2	87.4	48.2	89.9	47.2	89.2	46.3	88.8	89.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.0	19.8	17.3	67.2	93.9	6.4	0.9	62.6	94.5	58.9	95.1	55.3	95.6	50.6	96.3		
(5)			65.2	85.1	4.5	1.3	61.9	86.0	59.2	86.8	56.7	87.5	53.4	88.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	83.0	82.1	82.2	82.7	83.2	83.4	82.4	81.8	82.2	83.2	84.9	84.8	83.1		
	DBStd	1.87	1.66	1.58	1.43	1.45	1.65	1.66	1.54	1.30	1.40	1.54	2.09	1.83		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	12044	996	902	1015	996	1035	972	985	997	996	1081	1043	1026		
	CDD65	6569	531	482	550	546	570	522	520	532	546	616	593	561		
	CDH74	74048	5513	5004	5909	6179	6646	5803	5551	5733	6315	7749	7435	6211		
	CDH80	25973	1627	1495	1950	2184	2460	1869	1589	1675	2296	3406	3276	2146		
(14) Wind	WSAvg	6.5	7.1	6.0	4.0	4.3	5.8	6.8	8.4	9.7	8.6	7.7	5.1	4.7		
(15) Precipitation	PrecAvg	58.9	9.2	8.4	9.0	5.6	6.0	2.9	1.2	0.4	0.6	1.6	4.6	9.2		
	PrecMax	86.6	17.5	15.4	20.7	11.7	24.9	12.7	11.9	3.3	4.1	9.5	14.7	20.4		
	PrecMin	29.1	0.5	3.0	1.7	0.9	0.1	0.0	0.0	0.0	0.0	0.0	0.0	1.9		
	PrecStd	13.8	4.2	3.2	4.3	3.1	5.4	3.3	2.4	0.8	1.1	2.4	3.7	4.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.7	90.6	91.1	90.8	90.8	89.8	88.6	88.8	90.8	93.1	93.5	92.3		
		MCWB	80.7	82.0	81.9	81.9	81.6	80.5	78.8	79.1	80.4	80.9	82.2	81.5		
	2%	DB	89.3	89.3	89.9	89.8	89.8	88.8	87.7	87.9	89.8	91.8	92.4	90.6		
		MCWB	80.7	81.4	81.5	81.5	81.1	80.1	78.5	78.4	79.3	80.5	81.7	81.7		
	5%	DB	88.0	88.1	89.0	88.9	89.0	87.8	86.9	87.1	88.9	90.9	91.4	89.4		
		MCWB	80.6	80.7	81.2	81.2	80.7	79.8	78.2	77.9	78.8	80.4	81.3	81.3		
	10%	DB	86.7	86.9	87.8	87.9	88.0	86.8	86.1	86.3	88.0	89.9	90.2	88.0		
		MCWB	80.1	80.1	80.8	80.9	80.3	79.2	77.9	77.4	78.4	80.2	81.0	80.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.7	84.0	83.7	83.7	83.4	82.6	81.8	81.6	83.7	85.3	85.4	84.3		
		MCDB	87.6	89.1	89.3	89.2	88.7	87.7	86.2	87.3	89.4	91.3	91.9	90.1		
	2%	WB	81.9	82.3	82.6	82.7	82.4	81.4	80.3	80.2	81.8	83.5	83.6	82.8		
		MCDB	87.2	88.0	88.5	88.3	88.3	86.9	85.6	86.3	88.3	90.0	90.5	89.1		
	5%	WB	80.9	81.2	81.7	81.8	81.7	80.6	79.6	79.2	80.6	82.2	82.4	81.8		
		MCDB	86.5	86.8	87.5	87.5	87.4	86.3	85.0	85.5	87.2	89.1	89.4	88.0		
	10%	WB	80.1	80.4	80.9	81.1	81.0	79.9	78.8	78.4	79.5	81.0	81.4	80.9		
		MCDB	85.5	85.7	86.4	86.6	86.6	85.3	84.4	84.6	85.9	87.9	88.4	86.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	10.0	10.5	9.9	9.4	9.0	9.0	9.0	10.5	10.8	10.5	10.0		
		MCDBR	11.1	11.5	11.9	10.8	10.2	9.7	9.6	9.5	11.4	11.7	11.4	11.6		
	5% WB	MCWBR	5.4	5.7	6.0	5.2	4.7	4.5	4.6	4.6	5.4	5.0	5.1	5.6		
		MCWBR	10.4	11.0	11.2	10.4	9.9	9.3	9.4	9.0	10.7	10.8	10.8	11.1		
(40) Clear-Sky Solar Irradiance	taub		0.514	0.512	0.483	0.453	0.433	0.421	0.412	0.427	0.439	0.483	0.507	0.522		
		taud	2.173	2.184	2.266	2.359	2.423	2.458	2.473	2.405	2.352	2.228	2.177	2.141		
	Ebn at Noon		265	266	269	269	265	264	269	272	277	270	265	262		
		Edh at Noon	50	50	45	40	36	33	33	37	41	47	50	51		
(44) All-Sky Solar Radiation	RadAvg		1563	1652	1709	1702	1639	1594	1703	1903	2071	2039	1813	1508		
	RadStd		154	131	132	141	120	104	89	41	81	161	173	156		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	+0.43	+1.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page